

MARCEL KÜCK

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RESEARCH INTERESTS

Physical intelligence and embodied AI, with a particular focus on compliant and tendon-driven actuation and the coupling of learned control to soft, biological actuators. Interested in the design of soft hardware that mediates between engineered tissue and computational control, including soft bioreactors and mechanical stimulation systems for skeletal muscle tissue engineering. Prior work spans neural-network-based control of a tendon-driven musculoskeletal humanoid arm, parametric design of compliant hardware components, imitation learning for bimanual manipulation, and CAD-driven fabrication of soft and rigid components for humanoid platforms.

EDUCATION

Technical University of Munich, *Munich, Germany* 10.2019 – 05.2024
B.Sc. Computer Science. Specialisation in Robotics and Artificial Intelligence.

- Bachelor's thesis: *Implementation and Analysis of a Neural Network-Based Control Algorithm for Optimizing Joint-Space Trajectory Tracking for a Musculoskeletal Tendon-Driven Humanoid Robot Arm*. Addressed trajectory tracking on a redundantly actuated, compliant tendon-driven arm where rigid-body control assumptions break down. Conducted at Devanthro and supervised by the Chair of Robotics, Artificial Intelligence and Real-Time Systems (Prof. Alois Knoll).
- Member of TUM.AI, Software Development department.

University of Oxford, *Oxford, United Kingdom* 05.2024 – 07.2024
Machine Learning Summer School. Coursework and research seminars on reinforcement learning, generative AI, and machine learning in medicine. Selective admission (7%).

Google Machine Learning Bootcamp Europe 07.2022 – 11.2022
Three-month programme on applied machine learning. Selective admission (10%).

RESEARCH EXPERIENCE

Devanthro, *Munich, Germany* 03.2026 – Present
Affiliated Researcher

- Conducting research on control and embodied AI for Robody 5.0, the next-generation humanoid platform built on the OpenArm bimanual robotic arm.
- Developing an end-to-end pipeline from VR-based teleoperation through LeRobot policy training to on-hardware deployment.
- Investigating data-efficient imitation learning for bimanual manipulation.

Devanthro, *Munich, Germany* 10.2022 – 08.2024
Research Student and Founder's Associate

- Conducted research on neural-network-based control for tendon-driven humanoid arms (Robody 3.0), culminating in the bachelor's thesis on joint-space trajectory tracking for a musculoskeletal tendon-driven arm.
- Co-developed Robody 4.0 (Viper arm platform): contributed to mechanical assembly, performed Fusion 360 CAD work across structural and compliant components, and authored a Python script for parametric design of the elastic hull (the "skin") fitted over the rigid arm structure.
- Supported the founders across fundraising, strategy, and general operations as Founder's Associate.

INDUSTRY EXPERIENCE

MDK Engineering (marcelkueck.dev), *Munich, Germany* 01.2026 – Present
Founder

- Operating a solo AI and robotics consultancy providing AI integration and workflow automation to SMB and enterprise clients across multiple sectors.

- Designing and delivering retrieval-augmented generation systems, conversational agents, and AI-powered internal tools tailored to client domains.
- Building automation pipelines on n8n and Make to integrate LLM and ML capabilities into client business processes and operations.

Rechnungs-API (rechnungs-api.de), *Munich, Germany*
Co-Founder and CEO

02.2026 – Present

- Co-founded a B2B invoicing API; operating with a two-person team.
- Serving active paying customers with approximately 1,000 daily visits.
- Leading product direction, technical architecture, and go-to-market.

ShareYourSpace (shareyourspace.com), *Munich, Germany*
Co-Founder and COO

02.2025 – 01.2026

- Co-founded a marketplace for shared and flexible office space, an Airbnb-style platform for the coworking sector.
- Led day-to-day operational and product activities during the company's final year.
- Negotiated and executed the sale of the company and software platform to Pixida GmbH, a prior client.

Siemens AG, *Munich, Germany*
Software Engineer, Siemens SGI

09.2021 – 12.2023

- Designed and shipped a recommender system serving 50,000+ digital resources to 300,000+ users.
- Built machine-learning and web features across the SGI platform.
- Stack: Python, NumPy, TensorFlow, SQL, TypeScript, JavaScript, Angular.

PERSONAL PROJECTS

- **LeRobot SO-101 Manipulator:** End-to-end vision-language-action policy training and on-hardware deployment on the SO-101 arm using the LeRobot framework.
- **RustyRobots:** Open-source robotics framework for embodied AI in Rust; modular crates for perception, control, and simulation integration.
- **RustML:** Open-source machine-learning framework in Rust; tensor operations, automatic differentiation, and inference.
- **Roasterio:** Smart coffee-bean roaster combining a custom sensor stack with automatic roast-profile optimisation.
- **LaMarcello:** Retrofit controller converting classic espresso machines into smart, app-connected appliances; PID temperature control and flow profiling.
- **Murph:** Online medical consultation platform pairing medical students with elderly users for accessible primary advice.
- **Marie Lou Coffee:** Personal coffee-roasting venture; small-batch artisanal roasting, exploring a commercial launch.

TECHNICAL SKILLS

Programming:	Python, Rust, C++, TypeScript, JavaScript
Machine Learning & Robotics:	PyTorch, TensorFlow, LeRobot, ROS 2, MuJoCo
Fabrication & Prototyping:	Fusion 360, parametric CAD, SLA 3D printing, sensor integration, hardware-software integration
Tooling & Systems:	Linux, Git, Docker, NVIDIA Jetson, embedded systems

LANGUAGES

German (native) • English (fluent, C1–C2) • Spanish (fluent)